Agentic Architecture in Practice

Harness · LLM · Directives

A working model for how agentic systems are designed, governed, and extended — with **RockBot** as the example.

Rockford Lhotka · rockbot.dev · blog.lhotka.net

THE THESIS

What is an agent?

Not "an LLM with tools bolted on." In practice it's three distinct parts:

Agent = **Harness** + **LLM** + **Directives**

Harness

The code that controls execution — the loop.

LLM

Reasons and generates. A black box.

Directives

Shape behavior — scope, tone, rules.

Pull any one out and you don't have an agent.

The LLM is a black box

It's a microservice: **text in → text out.** Nothing more.

It doesn't remember the last call.

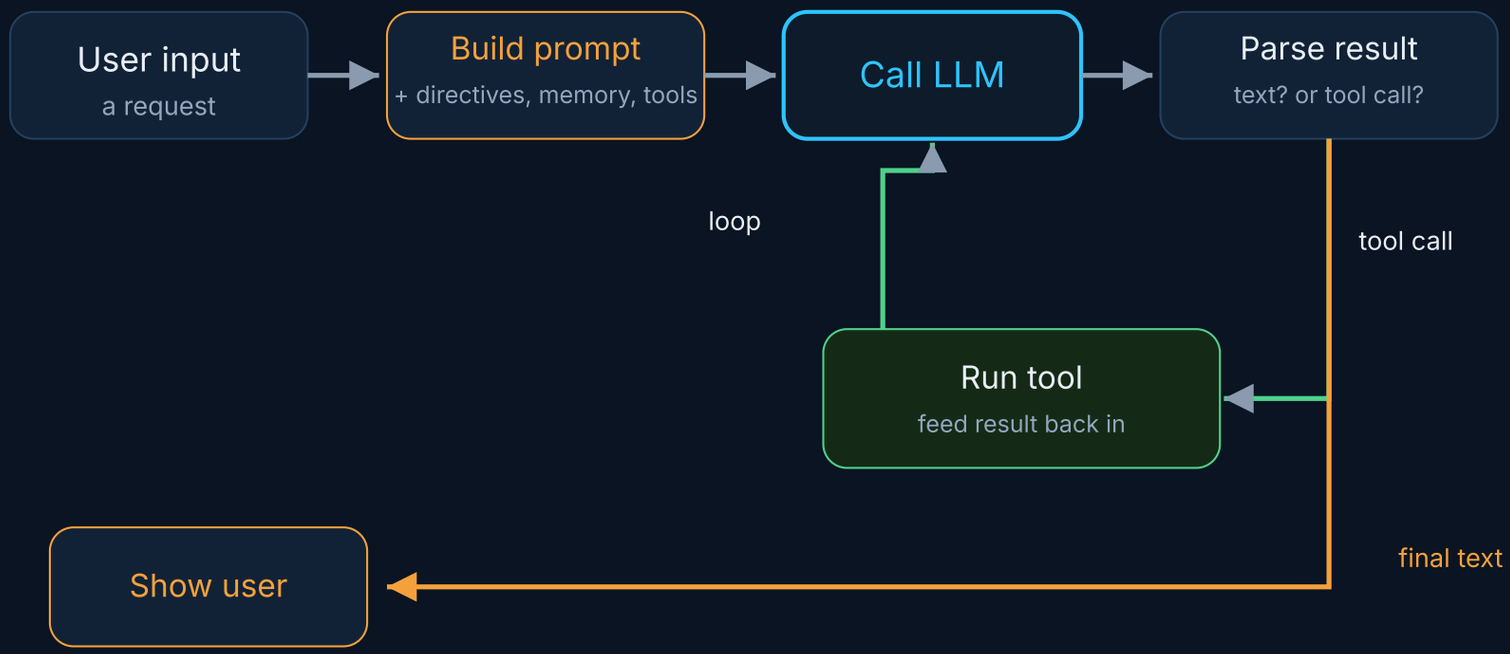
It can't call a tool — it just writes that it wants to.

It takes no action. It only generates text.

So where does all the "agent" behavior come from? →

The harness is the loop

This is where the real engineering lives.



The inner loop — call a tool, feed the result back, call the LLM again — is what makes an agent feel agentic.

PART 3 OF 3 — DIRECTIVES

Directives shape behavior

Scope · tone · rules-for-success. In RockBot, literally markdown files, composed into the system prompt.

📜 soul.md

Scope & identity.
Who the agent is, what it's for.

⚖️ directives.md

Rules.
Constraints that keep it safe and on-task.

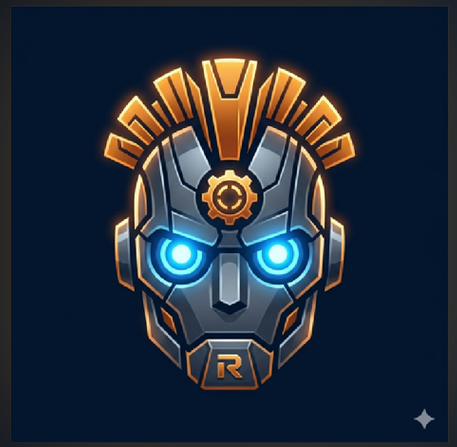
🎨 style.md

Tone.
How it communicates.

Layered & composable — base persona, then per-task, then per-skill:

soul + directives + style + task directive + skill → system prompt

Change behavior without changing code.



THE CONCRETE EXAMPLE

Meet RockBot

An **event-driven, process-isolated** autonomous agent framework for .NET.

"Nothing trusts the LLM."

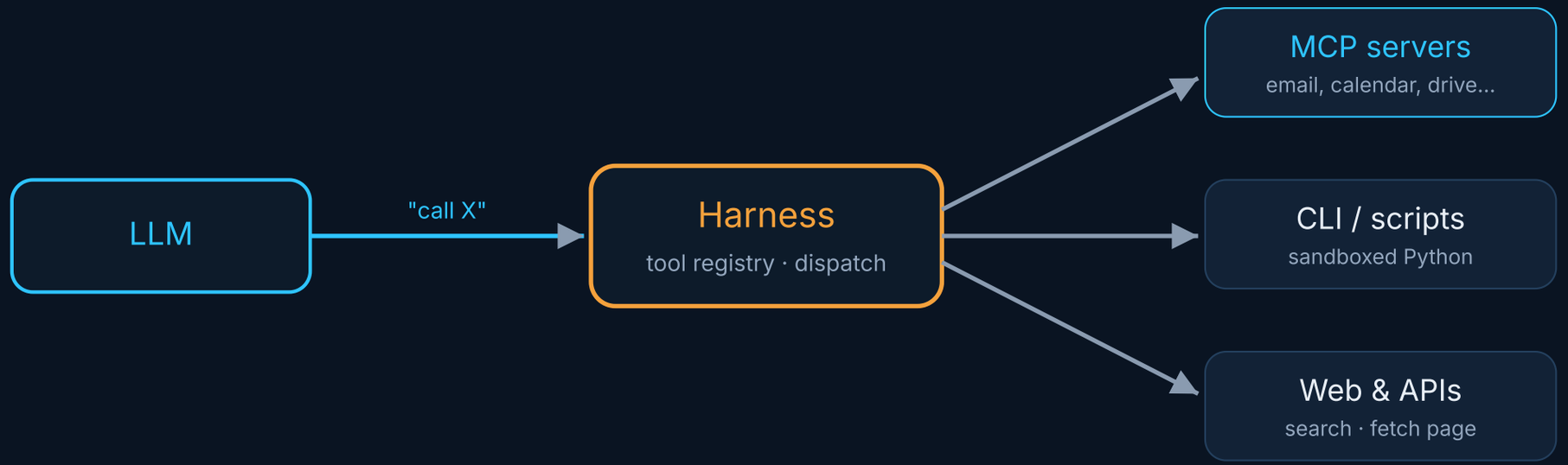
message bus (RabbitMQ) · isolated processes · least privilege · MIT · open source

Everything that follows is a real subsystem in this codebase.

REACHING OUTWARD

Tools: how the harness acts

The LLM only asks for a tool. The harness routes and runs it — then feeds the result back.

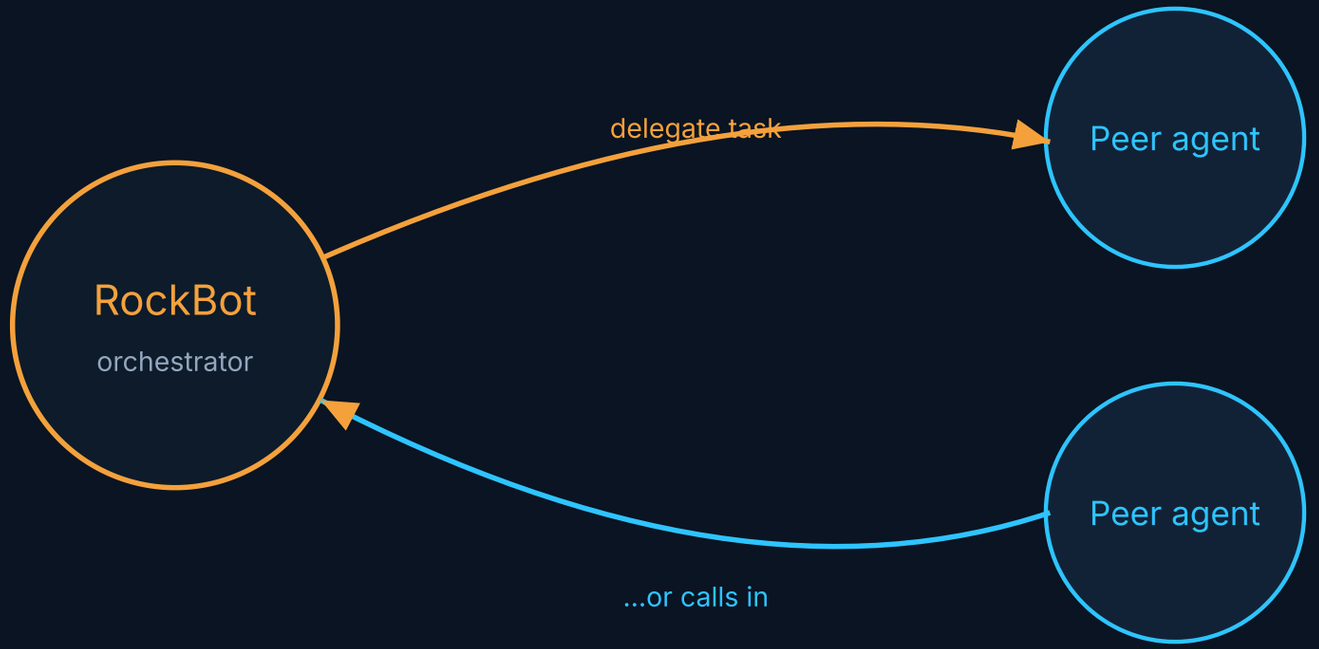


RockBot leans on **MCP** heavily — secrets live in the MCP servers, not the agent. (more on that later)

REACHING OUTWARD

Peers: agents calling agents

The **A2A** protocol lets RockBot delegate to other agents — or be delegated to.



A tool call reaches a service. An A2A call reaches a reasoning peer — its own harness, LLM, and directives.

THE PIVOT

The harness handles a lot

Once the core (loop + LLM + directives) is solid, this is what becomes possible:

The rest of the talk walks the spokes.

SPOKE 1 — CONTEXT THAT PERSISTS

Memory: three tiers + a graph

Conversation

Last N turns. Session-scoped, ephemeral. The here-and-now.

🗂️ Working

Shared, namespaced scratch space with a TTL. Hand-off between sessions, tasks & subagents.

🧠 Long-term

Permanent facts & preferences. Recalled per-turn by BM25 (+ vector) search.

Knowledge graph

Entities & relationships for structured, relational reasoning.

The trick isn't storing it — it's **injecting only what's relevant**, every turn, without token bloat.

Delta injection: a memory recalled once isn't re-injected the same session.

SPOKE 2 — REPEATABLE BEHAVIOR

Skills: fixed guides & evolving know-how

Memory stores facts. Skills store **how to do something** — markdown procedures, recalled by BM25.

📘 Subsystem guides

Authored, fixed. "How to use the email MCP," "how working memory works." Ship with the agent.

🌱 Mutable skills

The agent **writes its own** — captures a procedure that worked, then refines it over time.

The refine step happens offline — in the dream. →

SPOKE 3 — AUTONOMY

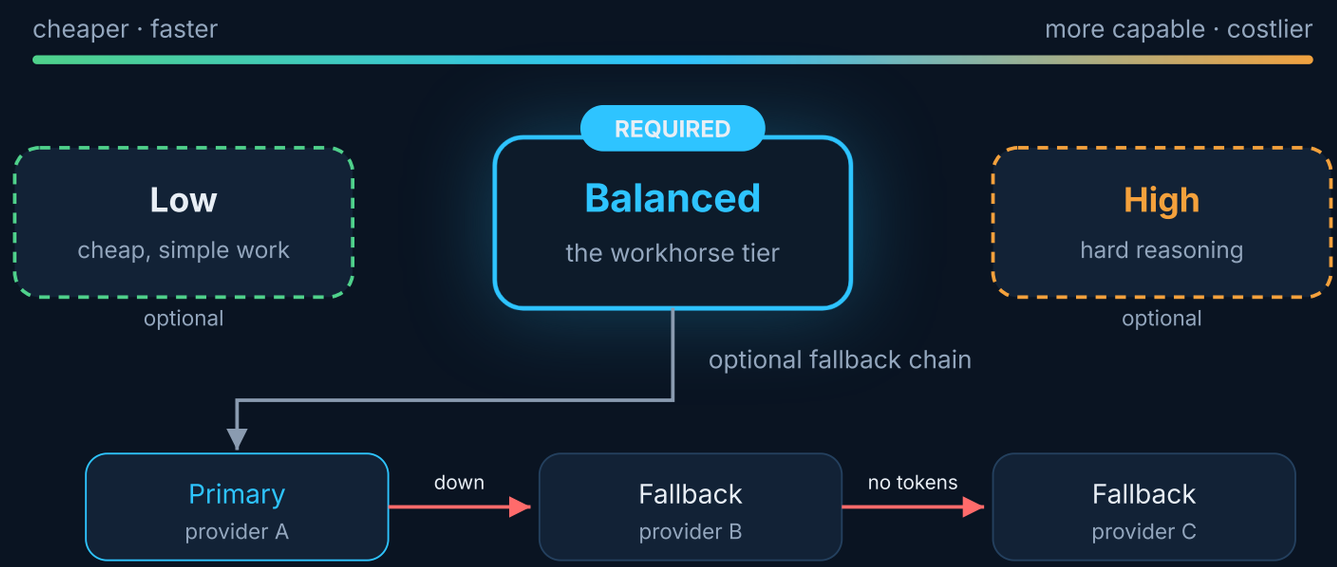
Tasks on a schedule

No user required. Cron-driven "patrol" tasks fire on their own — **each with its own directive.**

Same harness, same LLM — a different directive turns it into a different worker.

LLM routing: tiers & fallback

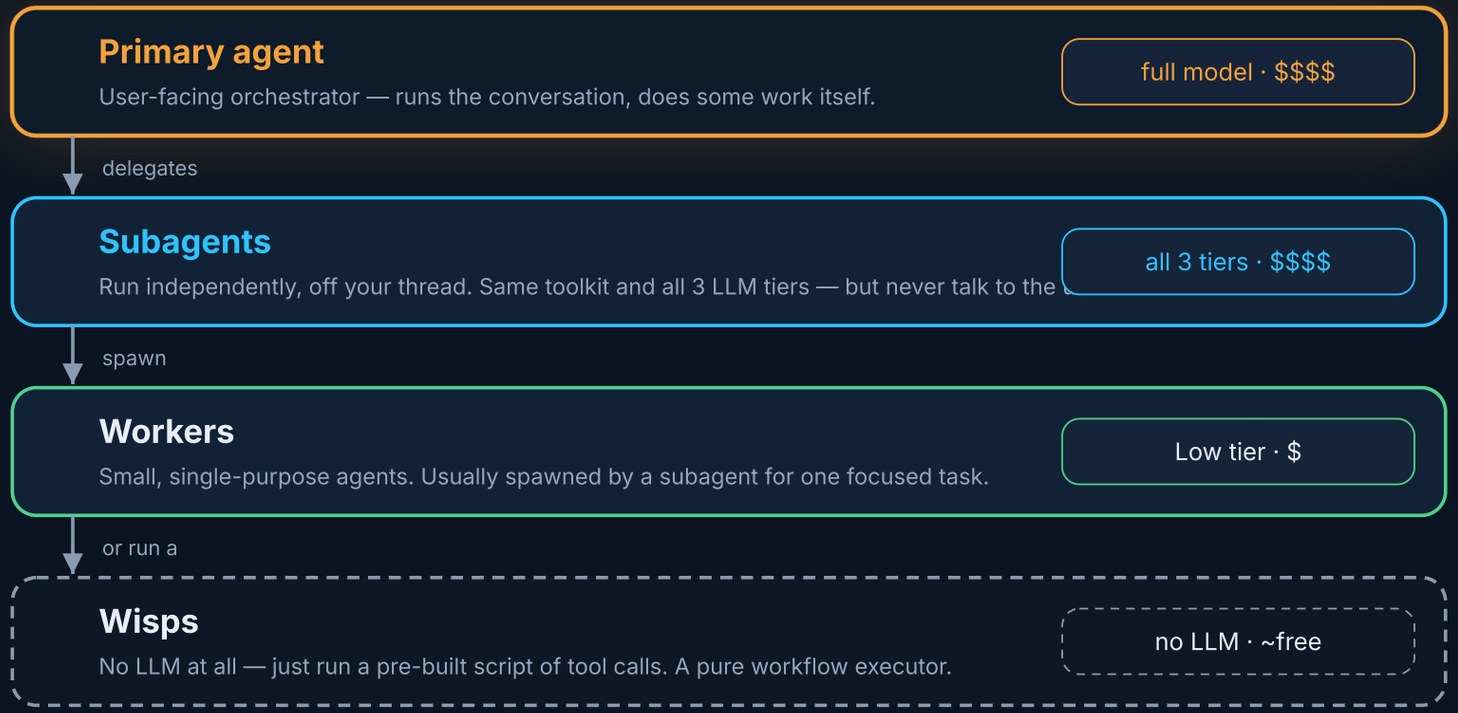
The LLM is swappable — so RockBot routes work across **cost/capability tiers**, and falls back when a provider fails.



Balanced is the one requirement. Low & High are levers for cost and capability — and the fallback chain keeps the agent working when a provider is down or out of tokens.

The orchestrator delegates down

The primary agent runs the conversation — and pushes each task to the **lightest executor that can do it.** Routine work drops to a cheap model, or no model at all.



Primary and subagents both reach for any tier. The savings come below: **workers** on the Low tier, **wisps** with no model at all.

SPOKE 5 — SELF-LEARNING

RockBot dreams

A background cycle that runs offline and **refactors the agent's own knowledge** — no user, no goals changed.

🧠 Consolidate memory

Merge duplicates, decay stale facts, mine anti-patterns from corrections.

🌱 Optimize skills

Refine, merge, and cluster the skills it has written.

🔀 Tune LLM routing

Learn which work belongs in which tier — Low, Balanced, or High — for the best cost/quality.

🔮 Infer preferences

Notice patterns across conversations; learn how you like to work.

Yesterday's experience makes tomorrow's agent sharper — automatically.

GOVERNANCE — TRUST & ISOLATION

Principle of least privilege

The core philosophy, restated: **nothing trusts the LLM.** The core agent holds no keys, no passwords.

A compromised prompt can't leak a secret the agent never had.

GOVERNANCE — TRUST & ISOLATION

Untrusted code runs sandboxed

RockBot can run Python and pull web pages — but never in-process.

Python execution

Runs in an **ephemeral, low-privilege container**. Spun up per run, destroyed after. No host access, no secrets, minimal blast radius.

Web search & fetch

Search the web and retrieve pages — but that content is **untrusted input**, handled with the same suspicion as the LLM's output.

Same pattern everywhere: **assume the LLM (and the web) can be wrong or hostile** — and contain it.

THE WORKING MODEL

Agent = **Harness** + **LLM** + **Directives**

Get the core right, and it extends:

memory · skills · LLM routing · self-learning · scheduled tasks · dreams · MCP · A2A · least privilege

From abstract "agent" talk to a working model for how agentic systems are **designed, governed, and extended.**

rockbot.dev · github.com/MarimerLLC/rockbot · blog.lhotka.net